

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4- Industry Problem Solving	Assessment:	Assessment Tool 2– Industry Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	T.Megha Harsha	Reg. No.:	192424389
Section / Batch:		Date:	

Code of Conduct

I T.Megha Harsha(192424389) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: MEGHA

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. A cybersecurity system historically detects 95% of malicious activities. After deploying a new detection algorithm, it correctly detects 194 out of 200 attacks. Test whether the new algorithm has significantly improved the detection rate using a p-value.

Given Data

Historical detection rate (p_0) = 95% = 0.95

Number of attacks tested (n) = 200

Correctly detected attacks = 194

Sample detection rate:

$$\hat{p} = 194/200$$

$$= 0.97$$

$$= 97\%$$

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

Null Hypothesis (H_0):

$$p = 0.95$$

Alternative Hypothesis (H_1):

$$p > 0.95$$

Step 2: Calculate Test Statistic

Formula:

$$Z = (\hat{p} - p_0) / \sqrt{[(p_0(1-p_0))/n]}$$

Substituting values:

$$Z = (0.97 - 0.95) / \sqrt{[(0.95 \times 0.05) / 200]}$$

$$= 0.02 / \sqrt{0.0002375}$$

$$= 0.02 / 0.01541$$

$$= \mathbf{1.298}$$

Step 3: Calculate p-value

For $Z = 1.298$

$$p\text{-value} \approx \mathbf{0.097}$$

Step 4: Compare p-value with α

$$p\text{-value} = 0.097$$

$$\alpha = 0.05$$

Since

$$0.097 > 0.05$$

Fail to Reject H_0

Final Conclusion

There is insufficient statistical evidence to conclude that the new cybersecurity algorithm significantly improves the detection rate beyond 95%.

2.A hospital introduces a new appointment scheduling system. Before implementation, the average patient waiting time was 42 minutes. After implementation, a sample of 64 patients has an average waiting time of 38 minutes with standard deviation 12 minutes. Test whether the new system significantly reduces waiting time using the p-value approach.

Given Data

Previous average waiting time (μ_0) = 42 minutes

Sample size (n) = 64

Sample mean ($\bar{x}$) = 38 minutes

Sample standard deviation (s) = 12 minutes

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

Null Hypothesis (H₀):

$$\mu = 42$$

Alternative Hypothesis (H₁):

$$\mu < 42$$

Step 2: Calculate Test Statistic

Formula:

$$Z = (\bar{x} - \mu) / (s / \sqrt{n})$$

Substituting values:

$$Z = (38 - 42) / (12 / \sqrt{64})$$

$$= (-4) / (12 / 8)$$

$$= (-4) / 1.5$$

$$= -2.667$$

Step 3: Calculate p-value

For $Z = -2.667$

p-value $\approx$ **0.0038**

Step 4: Compare p-value with α

p-value = 0.0038

$\alpha = 0.05$

Since

$0.0038 < 0.05$

Reject H_0

Final Conclusion

There is sufficient statistical evidence that the new appointment scheduling system significantly reduces patient waiting time.

3. A social media platform claims that the average daily usage time of its users is 3 hours. A random sample of 100 users has an average usage time of 3.2 hours with standard deviation 0.8 hours. Determine whether the data provide sufficient evidence to reject the company's claim at $\alpha = 0.05$.

Given Data

Claimed average usage time (μ_0) = 3 hours

Sample size (n) = 100

Sample mean ($\bar{x}$) = 3.2 hours

Sample standard deviation (s) = 0.8 hours

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

Null Hypothesis (H_0):

$$\mu = 3$$

Alternative Hypothesis (H_1):

$$\mu \neq 3$$

Step 2: Calculate Test Statistic

Formula:

$$Z = (\bar{x} - \mu) / (s / \sqrt{n})$$

Substituting values:

$$Z = (3.2 - 3) / (0.8 / \sqrt{100})$$

$$= 0.2 / 0.08$$

$$= 2.5$$

Step 3: Calculate p-value

Since it is a two-tailed test:

$$\text{p-value} = 2 \times P(Z > 2.5)$$

$$\approx 2 \times 0.0062$$

$$\approx \mathbf{0.0124}$$

Step 4: Compare p-value with α

$$\text{p-value} = 0.0124$$

$$\alpha = 0.05$$

Since

$$0.0124 < 0.05$$

Reject H_0

Final Conclusion

There is sufficient statistical evidence to conclude that the average daily usage time differs from the claimed 3 hours.

4. A data science team develops a new classification model. The existing model has an accuracy of 82%, while the new model achieves 86% accuracy on a randomly selected test sample of 500 observations. Formulate suitable null and alternative hypotheses and Hypothesis.

Given Data

$$\text{Existing model accuracy } (p_0) = 82\% = 0.82$$

$$\text{New model accuracy } (\hat{p}) = 86\% = 0.86$$

Sample size (n) = 500

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

Null Hypothesis (H_0):

$$p = 0.82$$

Alternative Hypothesis (H_1):

$$p > 0.82$$

Step 2: Calculate Test Statistic

Formula:

$$Z = (\hat{p} - p_0) / \sqrt{[(p_0(1-p_0))/n]}$$

Substituting values:

$$Z = (0.86 - 0.82) / \sqrt{[(0.82 \times 0.18) / 500]}$$

$$= 0.04 / \sqrt{0.0002952}$$

$$= 0.04 / 0.01719$$

$$= \mathbf{2.327}$$

Step 3: Calculate p-value

For $Z = 2.327$

p-value $\approx$ **0.0100**

Step 4: Compare p-value with α

p-value = 0.0100

$\alpha = 0.05$

Since

$0.0100 < 0.05$

Reject H_0

Final Conclusion

There is sufficient statistical evidence that the new machine learning model provides a significant improvement in classification accuracy.

5. A manufacturer claims that its production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective. Perform a hypothesis test at the 5% significance level using both (a) confidence interval and (b) p-value approaches. Compare the conclusions.

Given Data

Claimed defect rate (p_0) = 2% = 0.02

Sample size (n) = 2000

Defective products = 55

Sample defect rate:

$$\hat{p} = 55/2000$$

$$= 0.0275$$

$$= 2.75\%$$

Significance level:

$$\alpha = 0.05$$

(A) Confidence Interval Approach

Step 1: State the Hypotheses

Null Hypothesis (H_0):

$$p = 0.02$$

Alternative Hypothesis (H_1):

$$p \neq 0.02$$

Step 2: Calculate Standard Error

Formula:

$$SE = \sqrt{[(\hat{p}(1-\hat{p}))/n]}$$

$$SE = \sqrt{[(0.0275 \times 0.9725)/2000]}$$

$$SE = \sqrt{0.00001337}$$

$$SE = \mathbf{0.00366}$$

Step 3: Calculate Margin of Error

Formula:

$$ME = Z_{\alpha/2} \times SE$$

$$= 1.96 \times 0.00366$$

$$= \mathbf{0.00717}$$

Step 4: Construct 95% Confidence Interval

$$CI = \hat{p} \pm ME$$

$$= 0.0275 \pm 0.00717$$

$$\text{Lower Limit} = 0.02033$$

$$\text{Upper Limit} = 0.03467$$

Confidence Interval

(0.02033 , 0.03467)

or

(2.033% , 3.467%)

Step 5: Decision Using Confidence Interval

Claimed defect rate = 2%

The confidence interval indicates the true defect rate lies approximately between 2.03% and 3.47%.

Therefore,

Reject H_0

(B) p-value Approach

Step 6: Calculate Test Statistic

Formula:

$$Z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0)]/n}$$

Substituting values:

$$Z = (0.0275 - 0.02) / \sqrt{[(0.02 \times 0.98)/2000]}$$

$$= 0.0075 / 0.00313$$

$$= \mathbf{2.397}$$

Step 7: Calculate p-value

For $Z = 2.397$

Two-tailed p-value

≈ 0.0166

Step 8: Compare p-value with α

p-value = 0.0166

$\alpha = 0.05$

Since

$0.0166 < 0.05$

Reject H_0

Final Conclusion

Both the Confidence Interval approach and the p-value approach indicate that the actual defect rate is significantly different from the claimed 2%. Therefore, the manufacturer's quality claim is not supported by the sample data.

Summary Table

Problem	Test Statistic	p-value	Decision
Cybersecurity	1.298	0.0970	Fail to Reject H_0
Healthcare	-2.667	0.0038	Reject H_0
Social Media	2.500	0.0124	Reject H_0
Machine Learning	2.327	0.0100	Reject H_0
Manufacturing	2.397	0.0166	Reject H_0